The National University
Faculty of computer science
IT + CS

Mid Term

Linear Algebra and Geometry	Time: one hour

Answer all of the following questions

part A

Q «1» Find x, y, z, ω such that :

$$2\begin{bmatrix} x+y & 2z+\omega \\ x-y & z-\omega \end{bmatrix} = \begin{bmatrix} 6 & 14 \\ 2 & 10 \end{bmatrix}$$

Q «2» Given $A = \begin{bmatrix} -1 & 3 \\ 4 & -2 \\ 5 & 0 \end{bmatrix}$, $B = \begin{bmatrix} -3 & 2 \\ -4 & 1 \end{bmatrix}$

Find AB

Q «3» By using Sarrus' Rule, evaluate the following determinant

$$\det\begin{bmatrix} 3 & -1 & 2 \\ 0 & 2 & 1 \\ 4 & -4 & 1 \end{bmatrix}$$

part B

Q «4» Find the inverse of the matrix $\begin{bmatrix} -1 & -3 \\ 1 & 4 \end{bmatrix}$

Q «5» Solve the following system

$$x + y + z = 6$$
$$x - y + z = 2$$
$$-x - y + 2z = 3$$

By Gauss-Jordan and Cramer's Rule